RAJALAKSHMI ENGINEERING COLLEGE

Aravintth Murugesh S 250701068-2025-CSE A2

Started on	Saturday, 3 October 2026, 11:56 AM
State	Finished
Completed on	Saturday, 3 October 2026, 12:55 PM
Time taken	58 mins 25 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.

Example:

s1: ggtabe

s2: tgatasb

s1	a	g	**g**	**t**	**a**	**b**
s2	**g**	x	**t**	**x**	**a**	**y**	**b**

The length is 4

Solveing it using Dynamic Programming

For example:

Input	Result
aab azb	2

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
#include <string.h>

int main() {
    char a[100], b[100];
    scanf("%s %s", a, b);

    int n = strlen(a);
    int m = strlen(b);
    int dp[n + 1][m + 1];

    for (int i = 0; i <= n; i++)
        dp[i][0] = 0;

    for (int j = 0; j <= m; j++)
        dp[0][j] = 0;

    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= m; j++) {
            if (a[i - 1] == b[j - 1])
                dp[i][j] = dp[i - 1][j - 1] + 1;
            else
                dp[i][j] = dp[i - 1][j] > dp[i][j - 1]
                        ? dp[i - 1][j]
                        : dp[i][j - 1];
        }
    }

    printf("%d", dp[n][m]);

    return 0;
}
```

	Input	Expected	Got	
✔	aab azb	2	2	✔

	Input	Expected	Got	
✔	ABCD ABCD	4	4	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course